

Report on the conducted protein mobility simulation with and without covalent disulfide bonds between cysteines.

The main working files can be viewed in the [GitHub](#) repository.

Methodology and trajectory parameters.

First, I downloaded and visualized the protein structure in PyMOL. It should be mentioned that according to the CATH classification the protein belongs to 3.30.1350.10 (3 - Alpha Beta, 3.30 - 2-Layer Sandwich, 3.30.1350 - Crambin, 3.30.1350.10 - Thionin-like). The original pdb (and cif) file contained 7 amino acids (THR2, ILE7, VAL8, ARG10, SER22, ILE25, ILE34) in several alternative conformations (apparently for residues located on the protein surface, it is more difficult to unambiguously determine the position of side chain atoms due to their dynamics). For example:

ATOM	111	CB	AVAL	A	8	4.063	1.695	12.923	0.70	3.19	C
ATOM	112	CB	BVAL	A	8	4.269	1.433	13.215	0.30	9.52	C

Therefore, I reviewed the entire structure and deleted the duplicate atom coordinates to obtain plausible structures. An ethanol molecule was incorporated into the structure during protein crystallization, which I also removed since it is merely a solvent from which the crystal was obtained and has no relation to the protein structure itself. Thus, I saved the cleaned file as abcys.pdb for further work.

Since it was necessary to obtain the structure without disulfide bonds as well, I removed these bonds in abcys.pdb and added hydrogens to the sulfur atoms. After that, I saved the structure in the file abnocys.pdb (in fact, this might not have been necessary since PyMOL reads and displays these files identically, and the presence or absence of disulfide bonds can be determined at the topology creation stage using `gmx pdb2gmx`).

I assembled the topologies from the standard Gromacs force field library using AMBER99SB-ILDN for protein and AMBER94 for nucleic acids (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010) with the following command:

```
gmx pdb2gmx -f abcys.pdb -o abnocys.gro -p abnocys.top -i abnocys.itp -ss -ignh
```

The `-ss` flag allowed the creation of two different protein topologies with CYX (-S-S-) and CYS (-SH) cysteine residues.

Next, using the following sequence of commands, I created a water box with the proteins:

```
gmx editconf -f abnocys.gro -o indent -d 1.5
gmx solvate -cp indent.gro -cs spc216.gro -p abnocys.top -o solv.gro
```

Since the system has no net charge (when creating the topology: Total charge 0.000 e), there was no need to neutralize the charge by adding ions, so I proceeded to minimize and equilibrate the system (Figure 1):

```
gmx grompp -f enmin.mdp -c solv.gro -p abnocys.top -o abnocys_enmin
gmx mdrun -deffnm abnocys_enmin -ntmpi 1 -ntomp 8 -pin on -pinoffset 0 -pinstride 2
-update gpu -nb gpu -bonded gpu -pme gpu -notunepme -pmefft gpu -gpu_id 0 -v
```

The same steps were performed with eq_nvt.mdp and eq_npt.mdp.

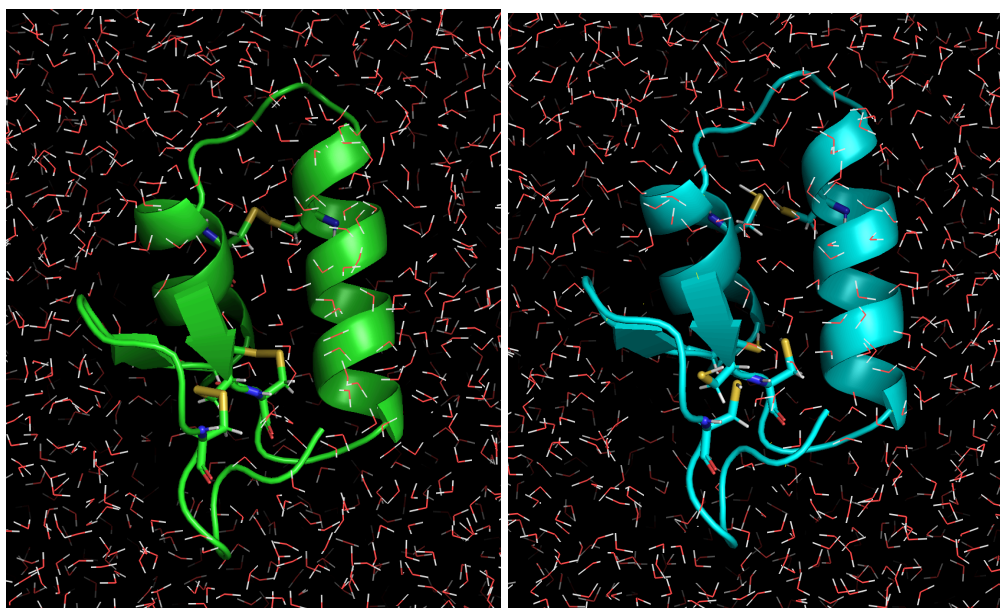


Figure 1. Visualization in PyMOL of structures (left - with disulfide bonds, right - without disulfide bonds) after energy minimization and NVT and NPT equilibrations.

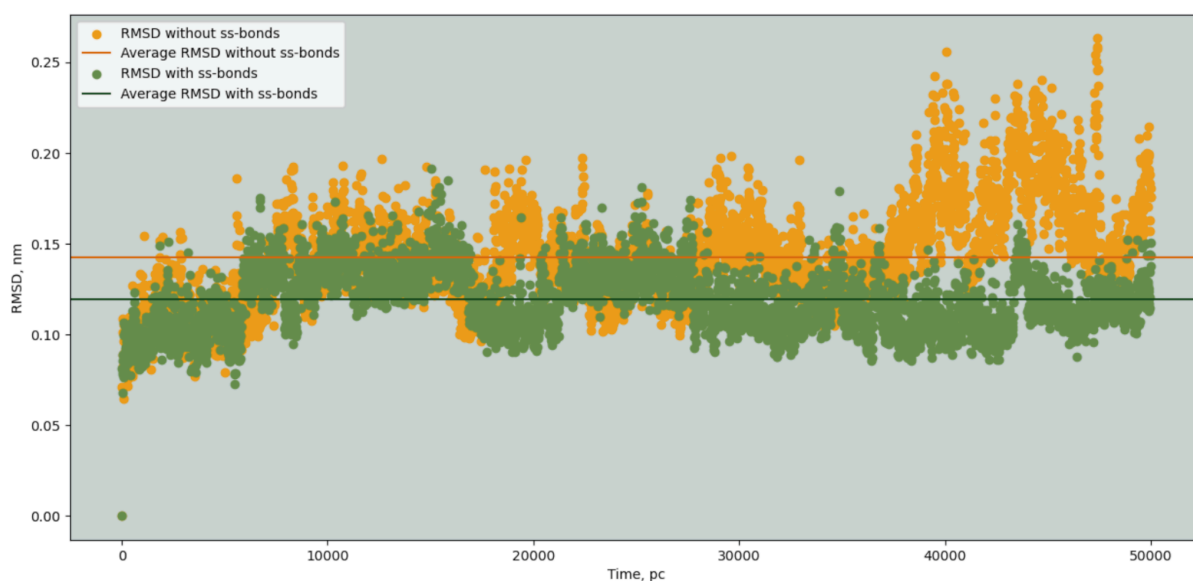
Thus, the systems for molecular dynamics simulation were ready.

The simulation duration was 50 ns (25,000,000 steps of 2 fs each). All files used to obtain the structures and trajectories can be viewed in the GitHub repository.

Trajectory Analysis

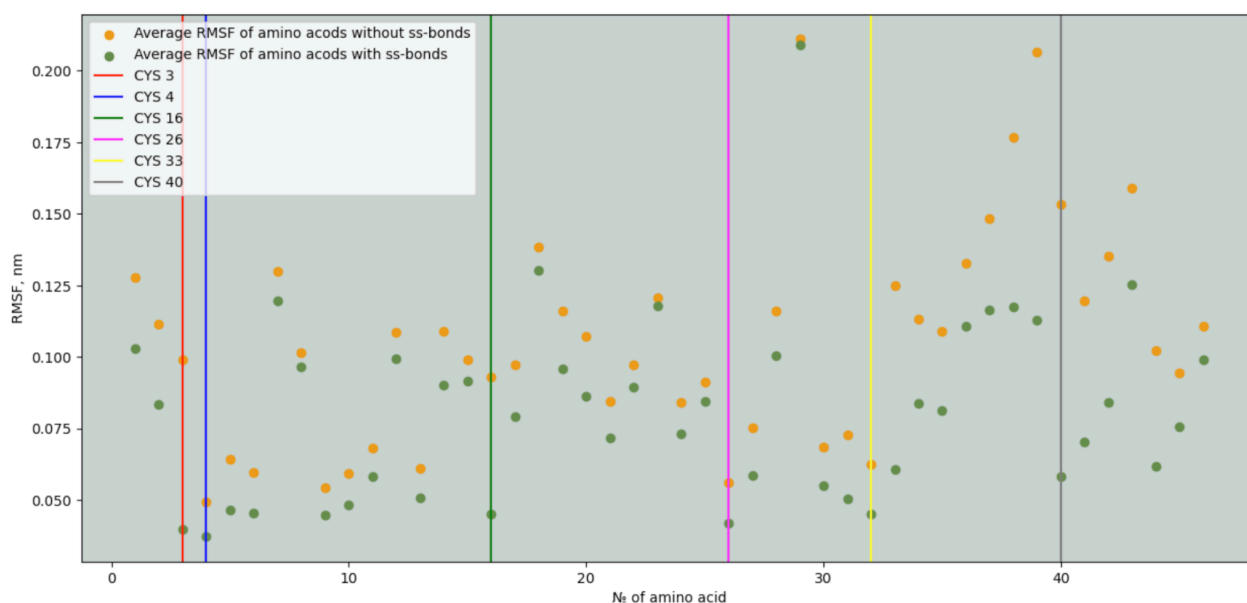
The mobility of proteins can be assessed by plotting graphs of the dependence of system parameters on the simulation time. Such parameters can include:

- root mean square deviation of structures from the initial (reference) structure along the trajectory – `gmx rms` (`-prev` for frame-by-frame comparison if necessary);
- root mean square fluctuation of atomic positions of amino acid residues from their initial positions along the trajectory – `gmx rmsf`.



Graph 1. Root mean square deviation (RMSD) of the structure from the initial one over time.

Graph 1 shows that, on average, the deviation of structures from the initial (reference) structure is greater in the case of the protein without disulfide bonds than with them.

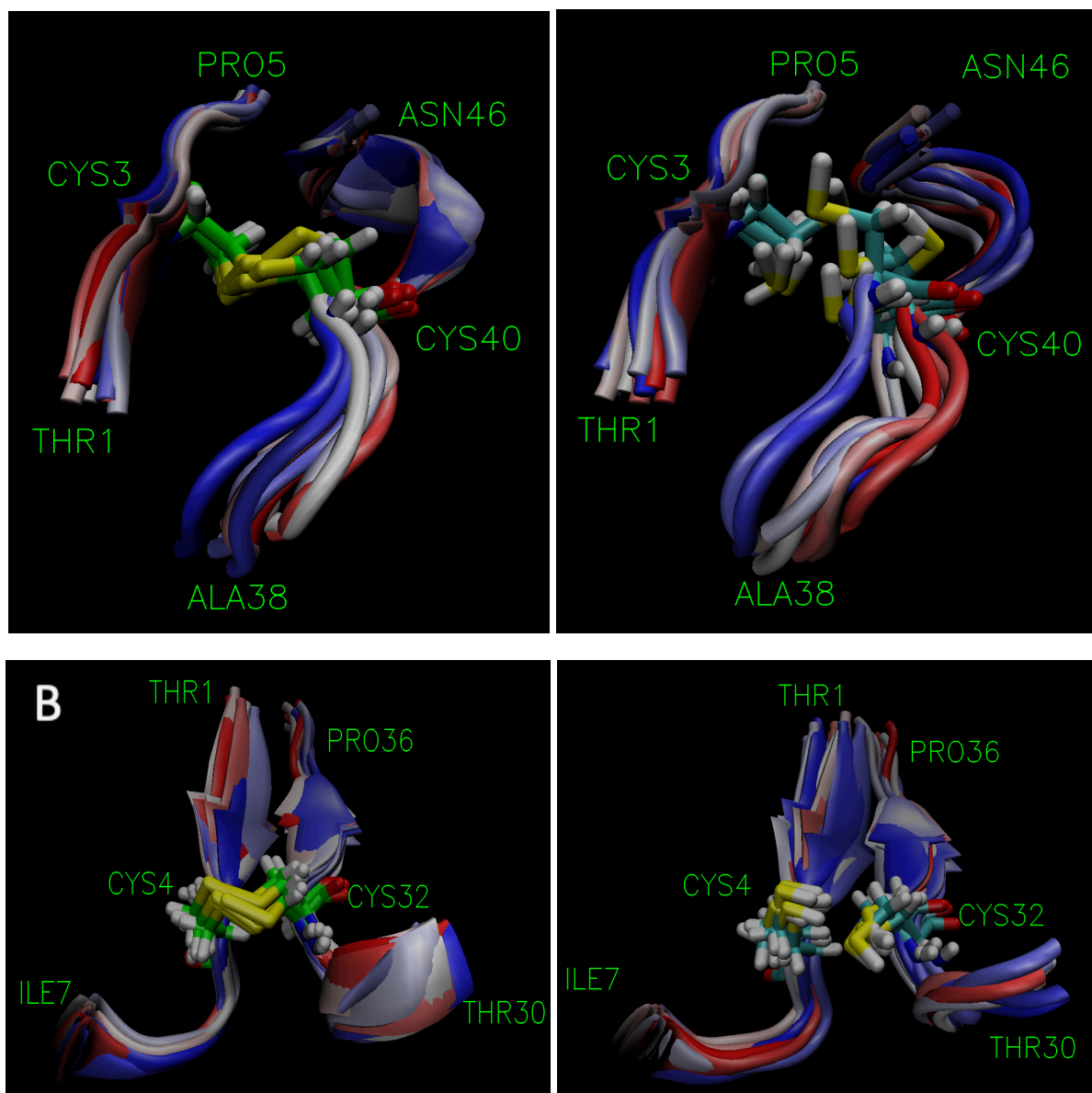


Graph 2. Root mean square fluctuation of atomic positions of amino acid residues from their initial positions over time. Vertical lines mark the cysteine residues.

According to Graph 2 all atomic positions of amino acid residues deviate from their initial positions during the trajectory, and the RMSF for residues of the protein without disulfide bonds is higher for each amino acid. The strongest deviation in positions is observed for the C-terminus of the protein since it is unstructured. However, this mobility increases significantly for the protein without the disulfide bond between CYS40 and CYS3 (Figure 2, A). Thus, it can be concluded that although the C-terminus has increased mobility compared

to other structural elements, its position is stabilized in the presence of S-S bonds, leading to significantly less mobility compared to the system without the disulfide bridge.

The behavior of cysteine residues in the two alpha-helices is interesting: the mobility of the helices relative to each other increases so much that the cysteine residues move away from each other and are at a distance of 7 Å (Figure 2, C, left, distance not shown). However, the position of cysteines in the beta-sheet is quite stable throughout the trajectory (Figure 2, B).



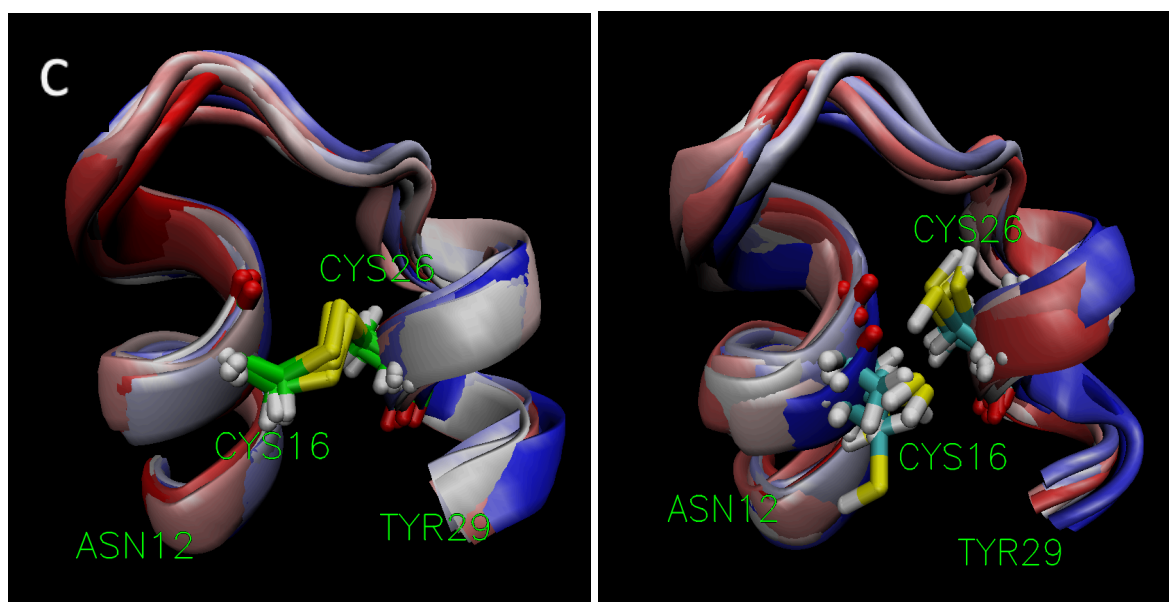


Figure 2. Visualization in VMD of every 10th frame of the trajectory in the case of the presence (left column) and absence (right column) of disulfide bonds. Polypeptide chains are colored depending on the frame sequence from most red, through white, to blue. Cysteine residues are shown for every 5th frame of the trajectory. A. Regions with amino acid residues 1-5 and 38-46 of the polypeptide chain. B. Regions with amino acid residues 1-7 and 30-36 of the polypeptide chain. C. Regions with amino acid residues 12-29 of the polypeptide chain. The figures for all regions show high mobility of the polypeptide chain in the absence of disulfide bonds, since the overlay of chains over time represents a more diffuse arrangement.

Thus, it can be concluded that the protein without disulfide bonds has greater mobility. The disulfide bonds act as specific structural anchors by maintaining the interatomic distance and orientation of the cysteine amino acid residue atoms, which in turn restricts the overall movement of the protein.